Book of Abstracts.
Students' International Conference "CERC" 2026
Volume 7, No. 1, ISSN: 2601-7989, ISSN-L: 2601-7989

Pseudo-DNA based Cryptography: Overview and Implementation

Moraru Andra, Nistor Augustin-Darius

Military Technical Academy "Ferdinand I"
Contact: augustin-darius.nistor@stud.mta.ro

Introduction: Cryptography is based on computationally hard problems that make deriving the plaintext back from the ciphertext impossible without the knowledge of a secret key. In 1994 Adleman proposed the idea of solving the problem of Hamiltonian path using deoxyribonucleic acid by leveraging its massive parallel computational and storage size that theoretically eases solving the problem [1]. The necessity of a more robust and easier to replicate algorithm, the paradigm of developing solutions for cryptography under laboratory conditions has changed to working with traditional computing means. Pseudo-DNA cryptography proposes just that, to add biological features and complexities into a digital system [2].

For a better understanding of the DNA construction and related advantages, some theoretical aspects are provided: the acid has a double-helical construction that consists of two pairs of complementary nucleotides, as defined by the principle discovered by Watson-Crick [3]. The complementarity of the nitrogenous bases can be defined as follows: Adenine (A) pairs with Thymine (T) and Cytosine (C) pairs with Guanine (G). Information encoding based on this newly created alphabet is done by selecting two bits in the original information and encoding them to their corresponding DNA base. The importance of the encoding scheme cannot be undervalued, because of the total possible twenty-four combinations of the binary encoding that result in a string of nucleotides, only eight of them are valid and that is because of the Watson-Crick principle of complementary bases.

Methods and Results: The current literature regarding DNA cryptography reaches a consensus regarding the encoding scheme and favors either a dynamic encoding scheme (the decision between the eight valid representations is chosen at runtime based on a random system [4]) or a static one. The main benefit of a dynamic encoding scheme is that it introduces confusion, meaning that data appears random to an attacker and, thus, he cannot perform further frequency analysis on it. Another key aspect discussed in most of the studies was the central operation that governs the whole algorithm. One of the most common approaches regarding operations is purely creating mathematical truth tables that are based on the complementarity discussed earlier. The underlying mechanism that revolves around performing additions, subtractions and XORs results in a base four truth table with the nucleoids as the truth value [4] [5] [6].

A second method used in performing cryptographic operations on DNA is the reproduction of Central Dogma of Molecular Biology at the level of digital code. The main operations regarding this method are splicing, transcription and translation of the DNA string and transforming them into corresponding ARN and protein that are then impossible to replicate without knowing the secret key parameters [7]. It is important to note here that the operation of splicing, transcription and translation represent procedures for permutations in the ciphertext that result in an increase in diffusion of the encryption algorithm.

From existing scientific literature, we selected fifteen papers that present studies that have been conducted in this cryptographic area. The selection of these papers was based on the

following criteria: studies must propose a pseudo-DNA algorithm for encryption, and have a section dedicated to analyzing either the security or the performance of their algorithm.

After analyzing these papers, the next step we took was to propose a method of categorizing the studies based on the following criteria: the assertion made on the security of the proposed algorithm, the theoretical space of the key used, the method for generating the keys, the encoding scheme type (static or dynamic), and the Shannon entropy value.

After the analysis was complete, we started working on a new algorithm that is derived from the proposals studied. This algorithm combines the efficacy of today's modern AES and key generation using PBKDF2. Also, we adopted the concept of using dynamic encoding for DNA at runtime based on the secret key. This feature of dynamic encoding was observed only in one of the fifteen studies we have gathered [4].

Conclusions: We conducted a literature survey and analyzed different papers proposed in this domain. Based on our findings we propose a C++ application that implements a hybrid AES and pseudo-DNA approach by using DNA as data storage and transmission. The application was then compared to other modern cryptography algorithms in terms of entropy and other relevant metrics.

Acknowledgements: Authors would like to thank to Major Eng. Ștefan-Ciprian Arseni for their guidance, insightful feedback, and encouragement throughout this project.

Keywords: *cryptography, information security, DNA encoding*

References:

[1] L. M. Adleman, "Molecular computation of solutions to combinatorial problems," *Science,* vol. 266, no. 5187, pp. 1021-1024, 1994.

[2] L. Y. S. Chu, "A review of DNA cryptography," *Intelligent Computing,* vol. 4, p. 0106, 2025.

[3] C. Yang, "Image encryption based on fractional chaotic pseudo-random number generator and DNA encryption method," *Nonlinear Dynamics,* vol. 109, no. 3, p. 2103–2127, 2022.

[4] M. Najaftorkaman and N. S. Kazazi, "A Method to Encrypt Information with DNA-Based Cryptography," *International Journal of Cyber-Security and Digital Forensics (IJCSDF),* vol. 4, no. 3, p. 417–426, 2015.

[5] H. Wen and Y. Lin, "Cryptanalyzing an Image Cipher Using Multiple Chaos and DNA Operations," *Journal of King Saud University – Computer and Information Sciences,* vol. 35, no. 7, 2023.

[6] A. Majumdar, "A Novel DNA-Inspired Encryption Strategy for Concealing Cloud Storage," *Frontiers of Computer Science,* vol. 15, no. 3, 2021.

[7] M. Mondal and K. S. Ray, "Review on DNA Cryptography," *arXiv preprint,* 2019.